

Non-Convex Polygon Packing via Simulated Annealing

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Abstract. The Problem addressed is to find the minimum length L^* of a square container Ω_L that can hold N Non Convex Polygons without overlap. Answering this exactly is NP-hard. We describe a practical solver combining two-phase Simulated Annealing (SA) with an outer bracketing-bisection loop that certifies the minimum side length L^* . For Collision detection we use the Separating Axis Theorem (SAT) constructing continuous energy function that depends on the penetration depth. We deployed embarrassingly parallel jobs on one node of the HPC cluster for $N \in \{5, 10, 25, 50, 100, 200\}$, with wall times from 3 minutes to 8 hours. Observed packing densities range from $\eta \approx 0.58$ at small N to $\eta \approx 0.48$ at $N = 200$. **Keywords.** polygon packing; simulated annealing; separating axis theorem; bracketing-bisection; embarrassingly parallel HPC; packing density scaling.

1. INTRODUCTION AND BACKGROUND

Take a fixed non-convex polygon. In our case we take a 15-vertex shape that was featured in an optimization Kaggle competition [?]. Make N identical copies. The problem is to find the smallest square that holds all of the N non-convex polygons without overlap. As N grows, the configuration space expands exponentially and exact enumeration becomes hopeless. Even for the circle packing of unit disks, the placement decision problem is NP-complete [?]. This is the usual way to probe a problem's computational complexity is NP: if the decision version is NP-complete, then the optimization version is NP-hard. Non Convexity adds a layer of difficulty making collision detection more expensive, and the search landscape more rugged, since it allows for interlocking of polygons. This added complexity is also reflected on their Minkowski sum characterizing pairwise overlap is itself non-convex and potentially disconnected.

Exact characterizations via the No-Fit Polygon (NFP), surveyed by Bennell and Oliveira [?], require convex decomposition for non-convex shapes and become impractical at the scales we study. Penalty-based SA consistently outperforms approaches restricted to strictly feasible states: as Leung et al. [2] demonstrated on orthogonal packing,

hard feasibility constraints fragment the search space into disconnected islands, while continuous penalties create traversable bridges between them. Genetic algorithms can explore structurally diverse configurations simultaneously [?], and Parallel Tempering [?] mixes exponentially faster than single-chain SA on multimodal landscapes, but both require substantially longer budgets to show advantages at our scales. As Sokal put it: “Monte Carlo is an extremely bad method; it should be used only when all alternative methods are worse.” That is the situation here.

2. PROBLEM FORMULATION

Let $\mathcal{P}$ be a fixed non-convex polygon with 15 vertices and area of the polygon $A_p = 0.245625$, calculated using the shoelace formula. To describe the placement of each copy of the polygon we use its centre coordinates (x_i, y_i) and orientation angle $\theta_i \in [0, 2\pi)$ for $i \in \{1, \dots, N\}$. A configuration is *feasible* if no two copies overlap and every copy lies within the square $\Omega_L = [-L/2, L/2]^2$. The goal is to minimize L . Specifically, the optimization problem is to find a feasible configuration of N copies in Ω_L that minimizes L for each N . In our case we will find feasible configurations for N in $\{5, 10, 25, 50, 100, 200\}$. Restricting search to non-overlapping configurations at all times confines the search to disconnected feasible islands in $\mathbb{R}^{2N} \times [0, 2\pi)^N$. We instead allow temporary, penalized violations that create navigable paths between those islands. This motivates a choice of collision detection that provides a continuous measure of overlap severity, not just a binary signal.

3. ALGORITHM

3.1. Collision Detection Pipeline

Collision detection is a key bottleneck in any packing and it is invoked after every proposed move to evaluate the energy change. The cost of naive pairwise overlap checks scales as $O(N^2)$. We use a spatial hash grid to only check neighboring polygons, and a multi-stage pipeline to reduce the number of SAT tests. The polygon is triangulated offline into $T = 13$ triangles, grouped into three clusters of sizes $(5, 5, 3)$. Detecting pairwise overlap naively costs $O(N^2)$ per iteration. A uniform spatial hash grid with cell size $h = 2r_{br} = 1.6$ restricts each query to a 5×5 neighbourhood of $\approx 261\eta$ candidates ($\eta = NA_p/L^2$); computing the energy change incrementally, touching only pairs involving the moved polygon, reduces the effective per-move cost to $O(1)$. Of the neighbour candidates, 96%

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are eliminated by polygon-level AABB checks; group-level and per-triangle AABB checks prune further before SAT is invoked.

SAT projects two triangles onto six edge-normal axes and computes the minimum overlap width $d \geq 0$ across all axes. If any axis shows a gap, the triangles are separated. When $d > 0$, it is the *penetration depth*: a continuous measure of how severely two polygons overlap. World-space vertex coordinates are projected directly onto each separating axis, avoiding any per-call recomputation of rotation matrices. This depth is what makes the energy approach tractable. A binary overlap test would assign equal cost to a barely-touching pair and a deeply embedded one, giving a flat energy surface with no gradient for SA to follow. The depth d is a directional signal, and feeding d^2 into the energy creates the continuous guidance that steers the algorithm toward feasibility from arbitrary initial configurations.

3.2. Energy Function and SA Schedule

The total energy of a configuration at container side L is

$$E = \lambda \sum_{i < j} \sum_{(a,b) \in \mathcal{T}_{ij}} d(a,b)^2 + \mu \sum_i \phi_i(L), \quad (1)$$

where $d(a,b)$ is the SAT penetration depth between triangle pair (a,b) , $\phi_i(L)$ is the squared AABB boundary violation of copy i , and (λ, μ) are penalty weights. Both terms are quadratic in the geometric violation, giving a smooth landscape that retains a directional gradient even for small overlaps. A configuration is declared feasible when the unweighted residual $\mathcal{F} = \sum_{i < j} (\text{overlap}_{ij}) + \sum_i \phi_i < 10^{-6}$.

Each SA trial is time-budgeted: temperature decreases geometrically, $T_k = T_0 \beta^k$ with $\beta = (T_{\text{end}}/T_0)^{1/K}$, from $T_0 = 1.0$ to $T_{\text{end}} = 10^{-5}$ over $K = 100,000$ proposals per trial. Step sizes (s_{xy}, s_θ) are adapted every 2,000 iterations: if the acceptance rate falls below 40% the steps shrink by factor 0.95; if it exceeds 60% they grow by 1.05. This couples step size to temperature so that high- T exploration uses large jumps and low- T refinement uses fine adjustments. The penalty weights (λ, μ) are held fixed within each trial; the caller supplies the starting values and may ramp them between trials to progressively enforce feasibility. Table 1 summarises the SA parameters.

3.3. Outer Minimisation: Bracketing, Bisection, and Polish

SA solves the feasibility problem at a fixed L ; an outer loop finds the minimum L . Starting from a grid layout with 15% padding, the algorithm shrinks L by 3% or grows by 5% per probe until it holds both a feasible upper bracket L_{high} and an infeasible lower bracket $L_{\text{low}} \geq L_{\text{area}}$. Binary search evaluates $L_{\text{mid}} = (L_{\text{low}} + L_{\text{high}})/2$ over 26 steps, reducing bracket width by $2^{-26} \approx 1.5 \times 10^{-8}$; a typical initial bracket of 0.25 certifies L^* to 3.7×10^{-9} in absolute terms. Before each SA run at a new L , polygon centres are rescaled by $\gamma = 0.98 \times L_{\text{new}}/L_{\text{old}}$, a warm-start that preserves structural information and reduces time-to-feasibility. After bisection,

Table 1. SA schedule parameters (per trial).

Parameter	Value
Proposals per trial K	100,000
T_{start}	1.0
T_{end}	10^{-5}
Temperature schedule	geometric ($\beta = (T_{\text{end}}/T_0)^{1/K}$)
Step adapt window	2,000 iters
Accept target band	[0.40, 0.60]
Step shrink / grow	0.95 / 1.05
Overlap power p	2 (quadratic, fixed)
Feasibility threshold	$\mathcal{F} < 10^{-6}$

an adaptive polish phase shrinks L by $\varepsilon \in [10^{-5}, 2 \times 10^{-3}]$ per attempt, targeting a 35% success rate over rolling windows of 20 probes, and runs until wall time expires. Every improvement is written to disk immediately; a global snapshot is flushed on **SIGTERM**, so SLURM preemption loses no progress. Each worker is seeded deterministically from (base seed, run ID, trial index) via SplitMix64 mixing, ensuring full reproducibility.

4. EXPERIMENTAL SETUP

All experiments ran on the University of Arizona HPC cluster (account **ece569**, **standard** partition). Production runs allocated five exclusive nodes per job, each with 32 cores and 16 GB RAM. Worker count per node was $W = 30$ (reserving two cores for the OS), giving $5 \times 30 = 150$ concurrent workers. Each node queued 64 seeds via **xargs -P 30**, so 320 seeds ran per job across three concurrency waves. Jobs for $N \in \{20, 50, 100\}$ used a 4-hour wall time (effective budget: 14,100 s after a 300-second post-processing cushion); the $N = 200$ job used 8 hours (28,500 s). Smoke tests at $N = 5$ and $N = 10$ used 3-minute wall times on the same configuration to validate the pipeline before committing to multi-hour allocations. SLURM sent **SIGTERM** 120 seconds before the hard kill, allowing the solver to flush its best snapshot.

A complementary set of single-node array jobs ran at most two seeds concurrently (8 tasks at $N = 25$, 6 at $N = 50$, 4 at $N = 100$, 2 at $N = 200$), with heavier trial budgets: 12 bracket and 10 bisection trials at $N = 25$, tapering to 8 and 6 at $N = 100$. No inter-worker communication occurred during any run; the global best was found in a post-hoc reduction pass over all output CSVs.

5. RESULTS

Table 2 reports the best feasible L^* and packing efficiency $\eta = NA_p/(L^*)^2$ for each instance size. The area lower bound $L_{\text{area}} = \sqrt{NA_p}$ is given for reference; $\eta = 1$ would require zero wasted space, which is geometrically impossible for this shape. “Runs” counts workers finding at least one feasible configuration.

Three findings stand out. First, the number of workers reaching feasibility drops sharply: from 15 of 15 at $N = 5$

Table 2. Best feasible side length, packing efficiency, and feasibility rate. Smoke tests ($N = 5, 10$) used 3-min wall time.

N	L_{area}	L^*	η	Runs	t_{first}
5	1.108	1.472	0.567	15	5 s
10	1.568	2.056	0.581	15	6 s
25	2.478	3.315	0.559	6	8 s
50	3.504	4.652	0.567	4	46 s
100	4.956	6.785	0.533	3	53 s
200	7.009	10.104	0.481	1	19 s

and $N = 10$, to 6 at $N = 25$, to a single run at $N = 200$. The fixed iteration budget $M = 105,000$ allocates M/N move proposals per polygon per trial, so at $N = 200$ each polygon is proposed 525 times per trial versus 4,200 times at $N = 25$. The algorithm under-explores a 600-dimensional space with a budget calibrated for 75 dimensions.

Second, best-of-best density η declines with N , from 0.58 to 0.48. Fitting $\eta(N) \approx \eta_\infty + c/\sqrt{N}$ to the $N = 50$ and $N = 200$ points gives $\eta_\infty \approx 0.40$ and $c \approx 1.22$. The scaling has a clean physical basis: wasted space attributable to boundary proximity scales with the container perimeter, which grows as $L \propto \sqrt{N}$, while bulk packing capacity scales with area, growing as $L^2 \propto N$; the relative boundary penalty therefore decays as $1/L \propto 1/\sqrt{N}$. The extrapolated η_∞ should be treated cautiously: the drop at large N reflects solver budget constraints as much as any geometric limit.

Third, median time to final best configuration was 205 seconds at $N = 25$ but over 3,250 seconds at $N = 100$, out of a 14,100-second budget. This places the best improvement at 23–26% into the run, confirming that the polish phase, not bisection, produces most of the gain at large N . Figures 1–5 show these dynamics in detail.

6. CONCLUSION

Choosing SAT over a binary overlap test was the decision that made this approach viable: penetration depth provides the continuous energy signal that steers SA toward feasibility from arbitrary configurations. The bracketing-bisection loop certifies L^* to sub-nanometer precision in 26 steps, and the adaptive polish phase productively fills whatever wall time remains. The experiments confirm the solver scales to $N = 200$ within practical HPC budgets, but reveal a clear bottleneck: the fixed iteration budget under-serves large N , and scaling $M \propto N$ is the first intervention to make. Future work has three natural directions. A CUDA implementation targeting the P100 (compute capability 6.0) could parallelise the SAT kernel across thread blocks, one block per polygon pair, exploiting the embarrassingly parallel multi-trial structure. Parallel Tempering, which runs replicas at a geometric temperature ladder and proposes Metropolis swaps between adjacent temperatures, could seed cold-chain refinement with configurations near deep feasibility basins, though it requires substantially longer budgets than tested here. A genetic algorithm treating full

layouts as chromosomes could maintain structural diversity across qualitatively different configurations simultaneously, a capability single-chain SA lacks.

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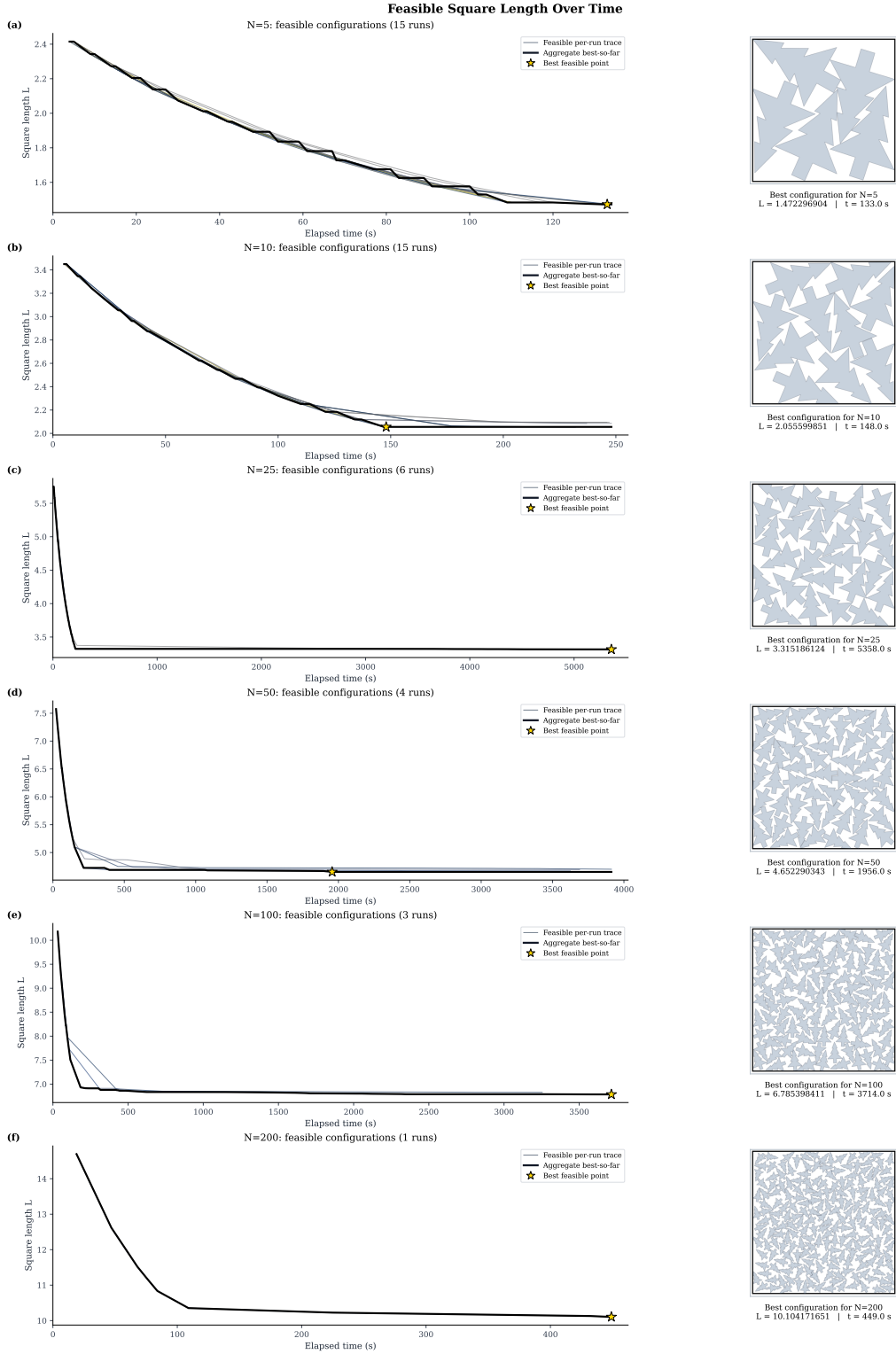


Figure 1. Feasible square side length L over wall time for $N \in \{5, 10, 25, 50, 100, 200\}$ (panels a–f). Thin lines are individual worker traces; the thick line is the aggregate best-so-far envelope across all workers. The right panel for each row shows the best packing geometry found: every copy is the same 15-vertex non-convex polygon placed inside Ω_{L^*} . At $N = 200$ only one worker reached feasibility, producing a single visible trace.

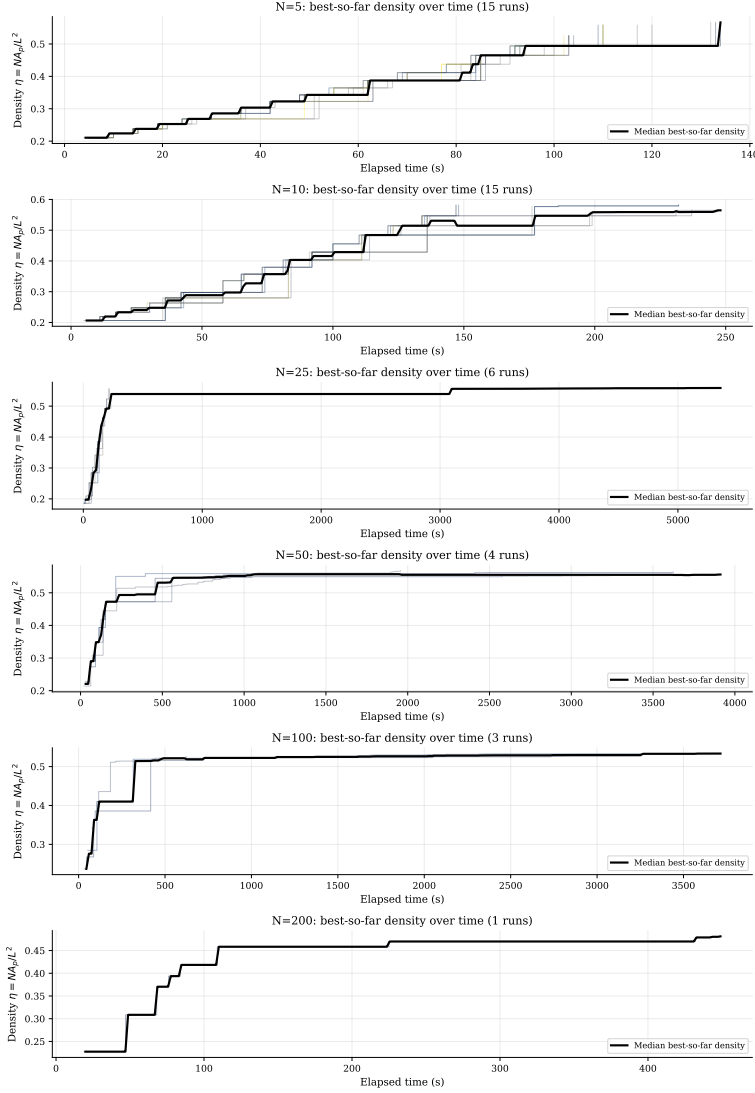


Figure 2. Best-so-far packing efficiency $\eta = NA_p/L^2$ versus wall time. Thin lines are individual workers; the thick line is the median across feasible workers. The plateau after initial improvement reflects the bisection phase locking in a bracket; gains thereafter come from the adaptive polish phase.

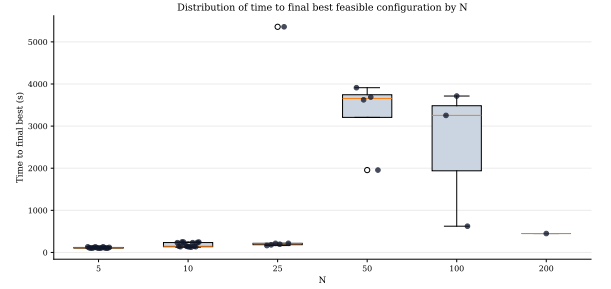


Figure 3. Wall time to final best feasible configuration by N . Boxes show the interquartile range; dots are individual workers. The sharp increase at $N \geq 50$ confirms that the polish phase, not bisection, produces most improvement at large scale.

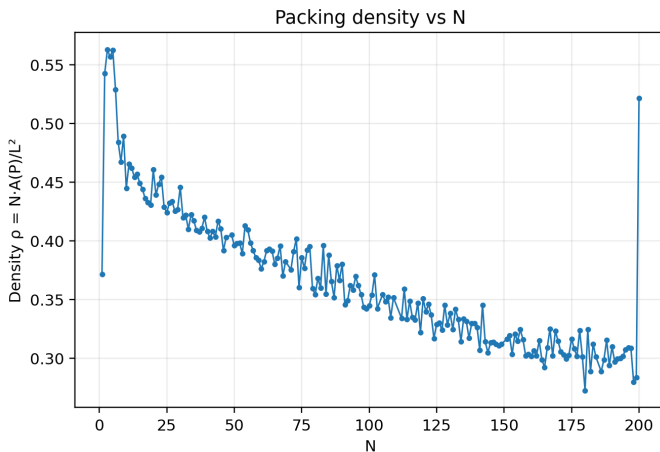


Figure 4. Best packing efficiency η versus N . Density declines monotonically, driven by both boundary geometry and the fixed iteration budget that under-explores larger configuration spaces.

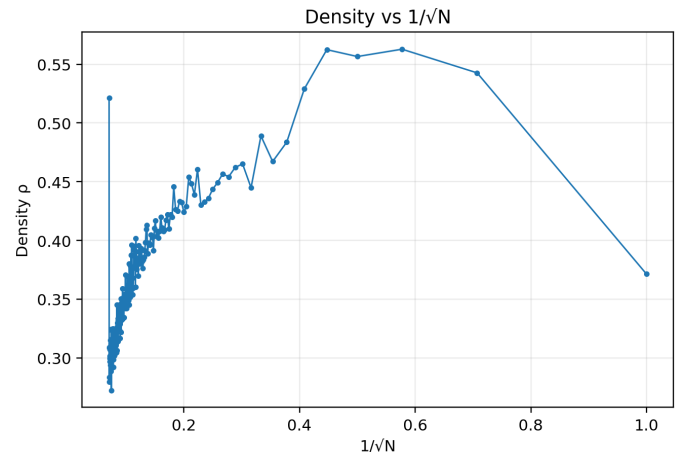


Figure 5. Packing efficiency η versus $1/\sqrt{N}$. The approximately linear trend supports $\eta(N) \approx \eta_\infty + c/\sqrt{N}$ with $\eta_\infty \approx 0.40$, consistent with the perimeter-to-area boundary argument.